

Data Analysis 2

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Begin by reading data into r

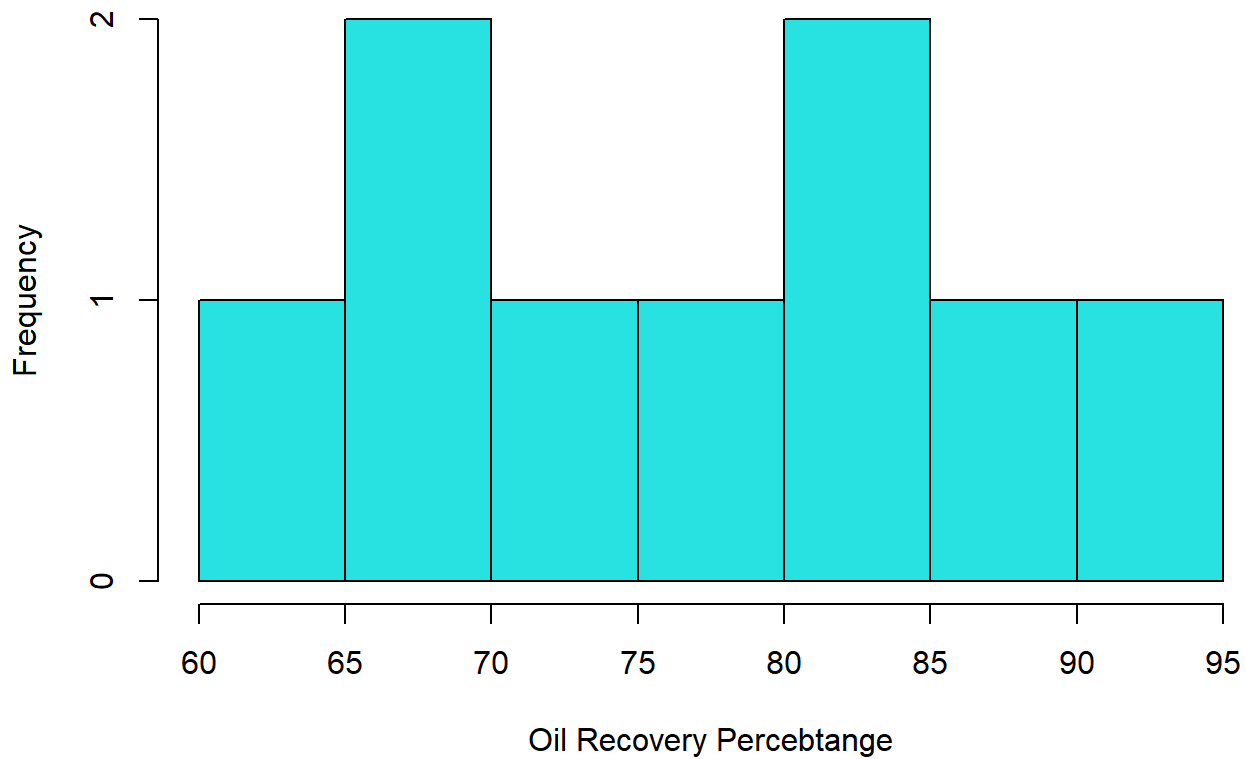
```
oildata <- read.csv("CRUDEOIL.csv",  
                    header = TRUE,  
                    sep = ",",  
                    stringsAsFactors = TRUE)  
  
head(oildata)
```

```
##  PRESSURE DIPANGLE OILREC  
## 1     1000         0 60.58  
## 2     1000        15 72.72  
## 3     1000        30 79.99  
## 4     1500         0 66.83  
## 5     1500        15 80.78  
## 6     1500        30 89.78
```

Question 1: Histogram for Oil Recovery

```
hist(oildata$OILREC, main = "Histogram of Oil Recovery", xlab = "Oil Recovery Percebtange", ylab  
     = "Frequency", col = 5)
```

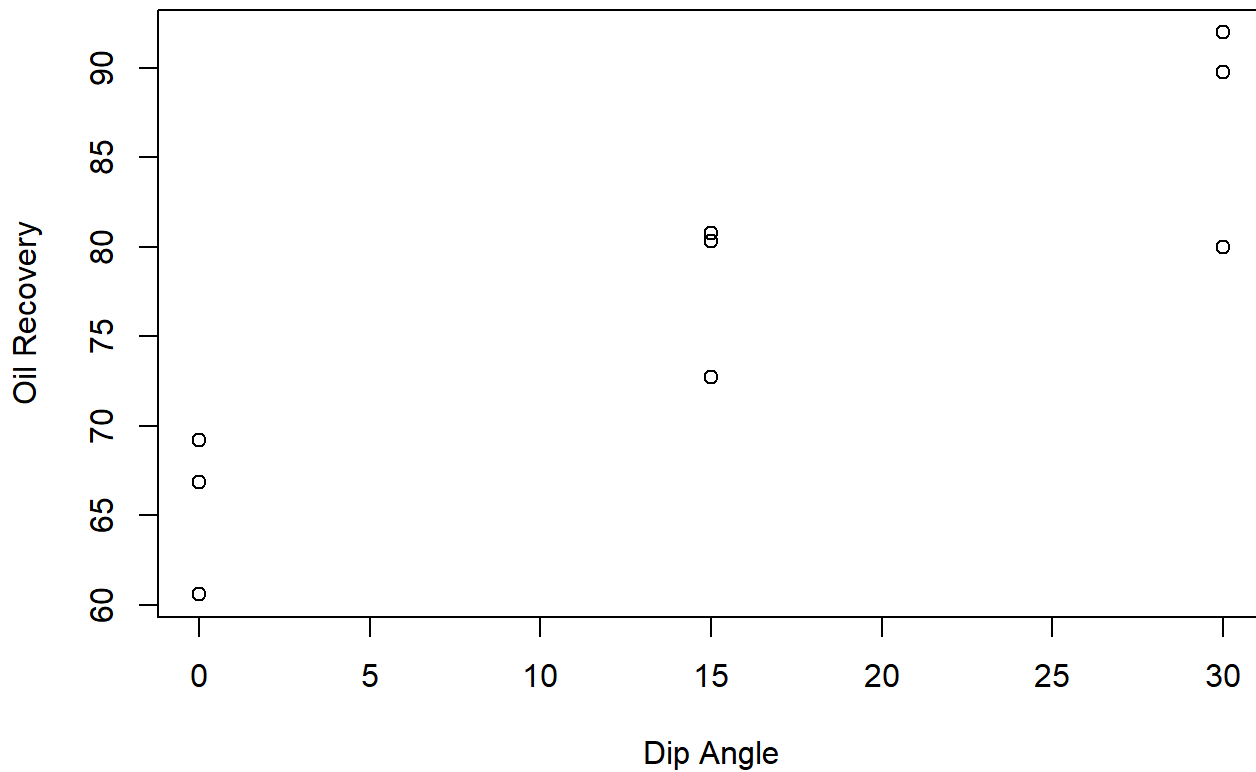
Histogram of Oil Recovery



Question 2: Scatterplot of Oil Recovery vs. Dipping Angle

```
plot(oildata$DIPANGLE,  
      oildata$OILREC,  
      xlab = "Dip Angle",  
      ylab = "Oil Recovery",  
      main = "Scatterplot of Oil Recovery vs. Dip Angle")
```

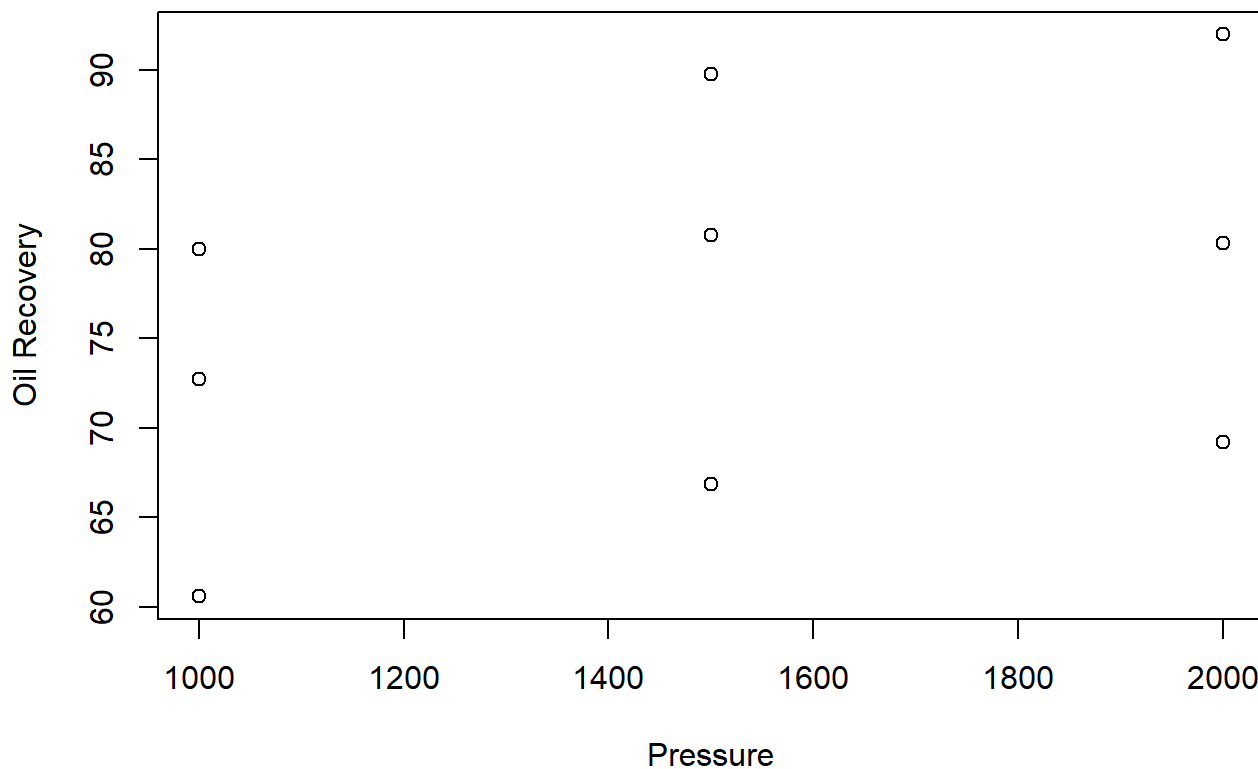
Scatterplot of Oil Recovery vs. Dip Angle



Question 3: Scatterplot of Oil Recovery vs. Pressure

```
plot(oildata$PRESSURE,  
      oildata$OILREC,  
      xlab = "Pressure",  
      ylab = "Oil Recovery",  
      main = "Scatterplot of Oil Recovery vs. Pressure")
```

Scatterplot of Oil Recovery vs. Pressure



Question 4: Center Variables and Give Names

```
oildata$PressCent <- oildata$PRESSURE - mean(oildata$PRESSURE)
oildata$DipCent <- oildata$DIPANGLE - mean(oildata$DIPANGLE)

head(oildata)
```

```
##  PRESSURE DIPANGLE OILREC PressCent DipCent
## 1    1000        0  60.58    -500     -15
## 2    1000       15  72.72    -500        0
## 3    1000       30  79.99    -500       15
## 4    1500        0  66.83        0     -15
## 5    1500       15  80.78        0        0
## 6    1500       30  89.78        0       15
```

Question 5: Fit the Second Order Model

```
oildata$PressCent2 <- oildata$PressCent^2
oildata$DipCent2 <- oildata$DipCent^2
oildata$PressDip <- oildata$PressCent * oildata$DipCent

oil.secondorder <- lm(OILREC~PressCent+PressDip+PressCent2+DipCent+DipCent2, data = oildata)

summary(oil.secondorder)
```

```
##
## Call:
## lm(formula = OILREC ~ PressCent + PressDip + PressCent2 + DipCent +
##     DipCent2, data = oildata)
##
## Residuals:
```

	1	2	3	4	5	6	7	8	9
	0.0100	0.5933	-0.6033	-0.9233	0.6200	0.3033	0.9133	-1.2133	0.3000

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.016e+01	9.069e-01	88.388	3.19e-06 ***
PressCent	9.397e-03	9.935e-04	9.458	0.002505 **
PressDip	1.133e-04	8.112e-05	1.397	0.256772
PressCent2	-1.334e-05	3.441e-06	-3.876	0.030400 *
DipCent	7.241e-01	3.312e-02	21.866	0.000209 ***
DipCent2	-6.867e-03	3.824e-03	-1.796	0.170407

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.217 on 3 degrees of freedom
## Multiple R-squared:  0.9949, Adjusted R-squared:  0.9865
## F-statistic: 117.6 on 5 and 3 DF,  p-value: 0.001223
```

```
round(summary(oil.secondorder)$coefficients[2],3)
```

```
## [1] 0.009
```

```
round(summary(oil.secondorder)$coefficients[5],3)
```

```
## [1] 0.724
```

Question 6: Test of Overall Significance

```
fstat <- (summary(oil.secondorder)$fstatistic)
```

```
fstat
```

```
## value numdf dendf
## 117.557 5.000 3.000
```

```
pf <- 1-pf(117.6,5,3)
round(pf,3)
```

```
## [1] 0.001
```

Question 7: Test of Significance for Interaction between Pressure and Dip Angle

```
summary(oil.secondorder)
```

```
##
## Call:
## lm(formula = OILREC ~ PressCent + PressDip + PressCent2 + DipCent +
##     DipCent2, data = oildata)
##
## Residuals:
##      1      2      3      4      5      6      7      8      9
## 0.0100 0.5933 -0.6033 -0.9233 0.6200 0.3033 0.9133 -1.2133 0.3000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 8.016e+01 9.069e-01 88.388 3.19e-06 ***
## PressCent    9.397e-03 9.935e-04 9.458 0.002505 **
## PressDip     1.133e-04 8.112e-05 1.397 0.256772
## PressCent2  -1.334e-05 3.441e-06 -3.876 0.030400 *
## DipCent      7.241e-01 3.312e-02 21.866 0.000209 ***
## DipCent2    -6.867e-03 3.824e-03 -1.796 0.170407
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.217 on 3 degrees of freedom
## Multiple R-squared: 0.9949, Adjusted R-squared: 0.9865
## F-statistic: 117.6 on 5 and 3 DF, p-value: 0.001223
```

```
anova(oil.secondorder)
```

```
## Analysis of Variance Table
##
## Response: OILREC
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## PressCent  1 132.45  132.45   89.4610 0.0025050 **
## PressDip   1   2.89    2.89    1.9521 0.2567721
## PressCent2  1  22.24   22.24   15.0251 0.0304003 *
## DipCent    1 707.85  707.85  478.1223 0.0002094 ***
## DipCent2   1   4.77    4.77    3.2246 0.1704073
## Residuals  3   4.44    1.48
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 9: Test of Significance for Second Order Dip Angle

```
tstat <- summary(oil.secondorder)$coefficients[6,3]
tstat
```

```
## [1] -1.795729
```

```
pvalue <- summary(oil.secondorder)$coefficients[6,4]
pvalue
```

```
## [1] 0.1704073
```

Question 11: Test of Significance for Second Order Pressure

```
sign.t <- round(summary(oil.secondorder)$coefficients[4,3],3)
sign.t
```

```
## [1] -3.876
```

```
psight <- round(summary(oil.secondorder)$coefficients[4,4],3)
psight
```

```
## [1] 0.03
```

Question 13: Removing Insignificant Variables

```
summary(oil.secondorder)
```

```
##
## Call:
## lm(formula = OILREC ~ PressCent + PressDip + PressCent2 + DipCent +
##     DipCent2, data = oildata)
##
## Residuals:
##      1      2      3      4      5      6      7      8      9
## 0.0100 0.5933 -0.6033 -0.9233 0.6200 0.3033 0.9133 -1.2133 0.3000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 8.016e+01 9.069e-01 88.388 3.19e-06 ***
## PressCent    9.397e-03 9.935e-04 9.458 0.002505 **
## PressDip     1.133e-04 8.112e-05 1.397 0.256772
## PressCent2  -1.334e-05 3.441e-06 -3.876 0.030400 *
## DipCent      7.241e-01 3.312e-02 21.866 0.000209 ***
## DipCent2    -6.867e-03 3.824e-03 -1.796 0.170407
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.217 on 3 degrees of freedom
## Multiple R-squared: 0.9949, Adjusted R-squared: 0.9865
## F-statistic: 117.6 on 5 and 3 DF, p-value: 0.001223
```

```
#Remove PressDip first
```

```
oil1 <- lm(OILREC~PressCent+PressCent2+DipCent+DipCent2, data = oildata)
summary(oil1)
```



```
##
## Call:
## lm(formula = OILREC ~ PressCent + PressCent2 + DipCent + DipCent2,
##     data = oildata)
##
## Residuals:
```

	1	2	3	4	5	6	7	8
	0.86000	0.59333	-1.45333	-0.92333	0.62000	0.30333	0.06333	-1.21333

```
##      9
## 1.15000
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.016e+01	1.009e+00	79.438	1.51e-07 ***
PressCent	9.397e-03	1.105e-03	8.501	0.00105 **
PressCent2	-1.334e-05	3.829e-06	-3.484	0.02527 *
DipCent	7.241e-01	3.685e-02	19.652	3.95e-05 ***
DipCent2	-6.867e-03	4.255e-03	-1.614	0.18185

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.354 on 4 degrees of freedom
## Multiple R-squared:  0.9916, Adjusted R-squared:  0.9832
## F-statistic: 118.3 on 4 and 4 DF,  p-value: 0.0002096
```

```
# Remove DipCent2
oilfinal <- lm(OILREC~PressCent+PressCent2+DipCent, data = oildata)
summary(oilfinal)
```

```
##
## Call:
## lm(formula = OILREC ~ PressCent + PressCent2 + DipCent, data = oildata)
##
## Residuals:
```

	1	2	3	4	5	6	7	8	9
	0.3450	1.6233	-1.9683	-1.4383	1.6500	-0.2117	-0.4517	-0.1833	0.6350

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.913e+01	8.984e-01	88.084	3.57e-09 ***
PressCent	9.397e-03	1.270e-03	7.396	0.000711 ***
PressCent2	-1.334e-05	4.401e-06	-3.031	0.029044 *
DipCent	7.241e-01	4.235e-02	17.099	1.25e-05 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.556 on 5 degrees of freedom
## Multiple R-squared:  0.9862, Adjusted R-squared:  0.9779
## F-statistic: 118.8 on 3 and 5 DF,  p-value: 4.568e-05
```

```
pressco <- round(summary(oilfinal)$coefficients[2,1],3)
pressco
```

```
## [1] 0.009
```

```
dipco <- round(summary(oilfinal)$coefficients[4,1],3)
dipco
```

```
## [1] 0.724
```

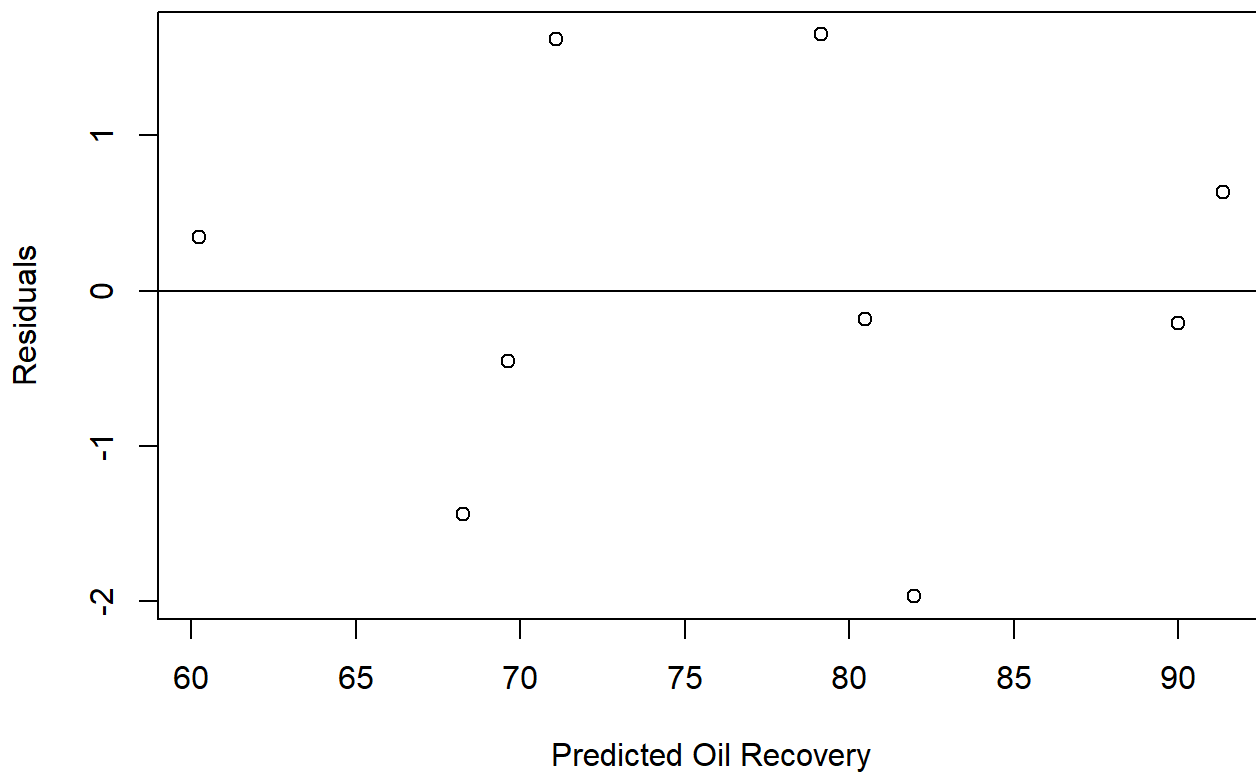
Question 14: Updated Residual Plot

```
oilfinalfit <- oilfinal$fitted.values
oil.res <- resid(oilfinal)

plot(oilfinalfit,
     oil.res,
     xlab = "Predicted Oil Recovery",
     ylab = "Residuals",
     main = "Residuals vs. Predicted Plot")

abline(0,0)
```

Residuals vs. Predicted Plot



Question 15: Updated Quantile Plot

```
qqnorm(oil.res)  
qqline(oil.res)
```

Normal Q-Q Plot

